

RESEARCH ARTICLE

Increasing the dosing accuracy of a screw dosing device by inline measurement of the product density

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Fine-powdered bulk materials exhibit high compressibility even under low pressure. The dosing error in volumetric dosing devices increases fundamentally with increasing product compressibility. In order to increase the dosing accuracy of a volumetric dosing of powdered bulk solids, the dosing process must be adapted to the product parameters. The most important parameter is the time-variable product density. With a pure volumetric dosage, it has a direct influence on the actually filled product quantity. Attempts are often made to obtain feedback by means of a scale in order to continuously adjust the volume. However, this is often associated with a delay. In this article, the approach is to measure the product mass flow directly by means of a microwave sensor and to regulate it to a predetermined mass flow. The experimental investigations were carried out with a screw dosing device FVS 3111. Two scenarios were tested: on the one hand, the integration of the sensor directly at the product delivery point, whereby the difference of the average value of the dosed quantity to the target weight could be improved with the sample product coffee; on the other, the sensor was integrated directly at the beginning of the screw and the remaining product flow was simulated. A reduction of the standard deviation compared with the conventional control could also be achieved, but here, the difference between the average of the dosed quantity and the nominal quantity is greater. The cause is presumed to be the too generalized assumption of product behaviour in the case of vertical conveying by means of a screw. In summary, the dosing accuracy of a screw dosing device could be increased by measuring and regulating the mass flow and thus the costs of a manufactured packaging could be reduced.

KEYWORDS

auger filler, control, dosing accuracy, microwave sensor, product density, screw dosing device, volumetric dosing

1 | INTRODUCTION

Screw dosing devices with vertical screw conveying are often used in filling machines in an industrial environment.¹ In particular for filling

powdery products such as coffee, flour, sugar, or milk powder, screw dosing devices are used in conjunction with vertical form, filling and sealing machines. In this case, the bulk material is conveyed from the silo (1), in the present case (Figure 1) via a transverse screw (3), into

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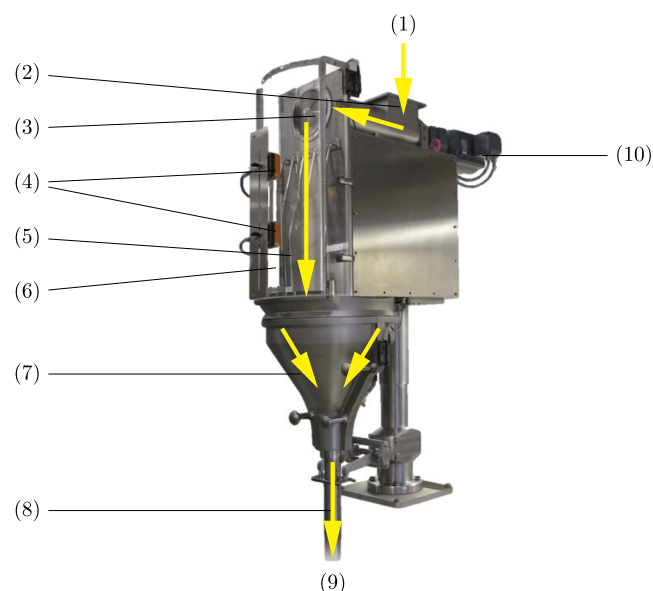


FIGURE 1 FVS auger filler² with indication of the product flow (yellow); (1) silo, (2) product feed, (3) transverse screw, (4) fill level sensors, (5) product fall, (6) filling container, (7) filling funnel including stirring devices, (8) dosing tube including dosing screw, (9) product discharge point, (10) engine of the transverse screw

the filling container (6) and passes via the filling funnel (7) into the dosing tube (8), where it is dosed by means of a dosing screw at the product discharge point (9).

Fine-powdered bulk materials show a high level of compressibility even under low pressure. This means that a constant reproducible density cannot be assumed under changing boundary conditions,¹ which is why stirring devices (7) are used for the comparability of the bulk density by loosening and to avoid bridging.^{1,3}

An example product mass flow depending on the filling level of the filling container is shown in Figure 2 for a powdery bulk material (coffee: Eduscho Gala Mild) at a constant screw speed of 100 rpm/min (assuming a filled screw and the dosing volume^{4,5}). The mass flow that does not remain constant is clearly visible due to the changing product density. With a constant volume to be dosed, the dosed quantity of product also changes or decreases with the increasing emptying of the filling container.

When filling products, legal requirements—for example, the prepackaging regulation⁶ in Germany—must also be complied with. The prepackaging regulation specifies the weight or volume of prepackages marked with the same nominal quantity and the tolerance of individual quantities. On the one hand, commercially marketed prepackages may be produced only in such a way that the filled quantity at the time of production does not fall below the nominal filling quantity on average across all prepackages and, on the other, the permissible number at which the defined permissible minus deviation exceeds the nominal filling quantity must be respected. If the dosing error is outside the required legal tolerance, the packages must be sorted out and laboriously returned to the cycle or disposed of expensively as waste.

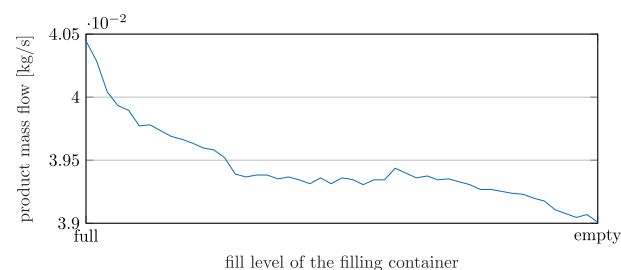


FIGURE 2 Exemplary mass flow over the emptying of the filling container—from full to empty—assuming a completely filled screw at a constant screw speed of 100 rpm; full corresponds to a filling volume of 0.055 m³ and empty of 0.0 m³ of the filling container

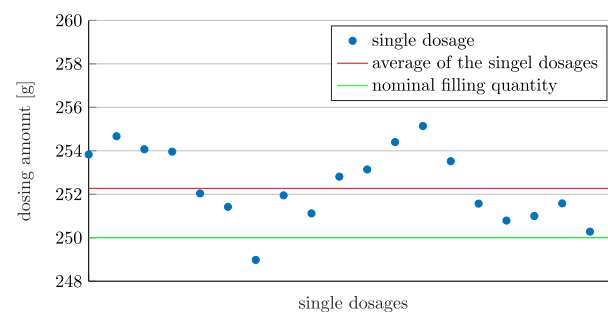


FIGURE 3 Average overfilling of a previous auger filler for the product coffee

In order not to violate the legal regulation due to the high compressibility of fine-powdered bulk goods with low pressure load, this leads among other problems in the industrial environment to an overfilling of the prepackages. This is shown as an example in Figure 3 for filling coffee with a nominal filling quantity of 250 g in 20 individual doses.

According to previous investigations, the level of overfilling is process-dependent (from the machine settings) and product-dependent and shows a variation of between 0.8 and 7.5% of the filled product quantity,⁴ whereby the deviation is greater for more flowable product. In the present case, the mean dosage is 252.27 g, which corresponds to an overfill of 2.27 g or 0.89%. The course over the 20 individual dosages shows an acceptable periodic course which, according to Vetter,¹ is caused by the intermittent filling of the screw filling device and the pulsating delivery, which is also dependent on the geometry of the dosing tube. In practice, especially when the machine is started, powdery products lead to increased rejects or an overfill or underfill of the nominal filling quantity due to the flowability and the solidification of the bulk material over time.⁷ According to investigations, this dosing error depends not only on the density of the material but also proportionally on the error of the step angle.⁸

In order to minimize the dosing error and thus the dispersion of the dosing quantities as much as possible and to reduce overfilling—a cost-saving potential for the manufacturer—some approaches are known from the literature. Many investigations aim to determine

dependencies between product and process parameters and thus to optimally operate the dosing device depending on the product and to increase the accuracy.^{9,10} For example, maximizing the dosing time also reduces the dosing error in many packaging goods, whereby product can also get into the sealing seam and lead to leaks.^{4,11} Furthermore, approaches are also known in which the machine design is optimally designed by means of simulations depending on the products to be filled.^{10,12} In most cases of powdered products, attempts are made to keep the mass flow as constant as possible by installing mechanical elements. In addition to the installation of stirring and special versions thereof,¹³ the use of level sensors—fixed in or onto the filling container—in order to keep the cross screw and thus the filling container at an almost constant level, increase dosing accuracy. The dosing accuracy can also be increased, according to Kammler and Baur,¹⁴ by using a number of screws arranged one after the other, in particular one for the coarse dosing and another with a lower nominal volume for the fine dosing. In practice, attempts are often made to obtain a direct feedback of the filled quantity through the installation of scales and to control the dosing screw. Often, however, a direct measurement of the product quantity is not possible at the delivery point, which is why attempts are made to continuously adjust the volume or the number of revolutions of a dosing screw by using optimization and statistical methods, which on the one hand lead to delays and on the other are approaches heavily reliant on the statistical methods used, such as averaging. A method for weighing the filled product using a flap scale¹⁵ is also known from the literature. Here, the product is dosed into a tubular bag by means of a dosing device, which is positioned on a flap scale with the feedback being fed to the controller. After the end of the dosage, the flaps (scale) open and the package is dispensed. If the positioning is inaccurate or the hose is not completely on the scale, this can lead to significant deviations.

Furthermore, Blagoveshchenskaya et al.¹⁶ describe an approach to using a neural network model to control the process of dosing bulk materials, in which control parameters are adapted by means of a neural network, since the control of a dosing device by means of a neural network controller cannot be used or is impossible for powdery products because control errors for impacts are not predicted. For this system, the learning error was 1.04%. However, the authors did not provide any information on the process parameters used and the resulting dosing error.¹⁶

In summary, the conventional methods and regulations for the volumetric dosing of finely powdered bulk materials are not optimal and offer a great potential for reducing the costs of a product's packaging.

The aim is therefore to find a way to develop a dosing process that is automatically adapted to the product parameters—especially the fluctuating product density—in order to increase the dosing accuracy of a volumetric dosing of powdered bulk materials. The relevant product parameters, in this case the density, are to be determined by means of a sensor during dosing. With the help of the temporal density values and the knowledge of the temporal volume flow, the actual temporal mass flow should be determined.

The determined actual mass flow is then to be controlled to a specified target mass flow by means of a model-based control. The target mass flow is determined or planned from the product quantity to be dosed and the available dosing time. By using a direct feedback over the already dosed product quantity, the dosing accuracy (minimizing the scattering of the individual dosed quantities) is to be reduced. This saves product and reduces the cost of manufactured packages.

2 | PRODUCT CHARACTERIZATION

In this study, coffee (coffee: Eduscho Gala Mild—Tschibo GmbH) is used as an example product, as coffee is often industrially dosed by means of a vertical screw dosing device.

In order to accurately describe the dosing of powders, it is necessary to analyse the relevant bulk material properties. Critical product attributes must be compatible with the process parameters in order to achieve strong productivity with high dosing accuracy of a vertical screw dosing system. Due to the large number of factors regarding the dosing accuracy, the first step in this study is to characterize the bulk material used. Table 1 shows an overview of the possible material properties to be analysed. In this paper, the focus is mainly on the bulk density, the tapped density, the resulting Carr's index, the angle of repose, and the shear properties, especially the flow function.

2.1 | Bulk density and tapped density

The bulk density generally represents the average density of the powder sample—that is, the volume of a bulk material depending on the size and shape of its particles. In order to determine the bulk density, a measuring cylinder with a capacity of 100 ml was filled and the density was determined using a jolting volumeter (Jel STAV II from

TABLE 1 Summary of the quantitative bulk material properties and the corresponding material attributes

Bulk properties	Material attributes
Particle size characterization	Particle size distribution
	Aspect ratio
	roundness
Bulk properties	Bulk density
	Tapped density
	True density
Flow properties	Carr's index
	Angle of repose
Shear properties	Flow function
	cohesion
	Angle of internal friction
	Wall friction angle

J. Engelsmann AG, based on DIN ISO 697). The tapped density was then compacted by compacting the powder bed with 1250 tamping movements in accordance with DIN EN ISO 787-11. The bulk density was 320.33 kg/m³ and the tapped density 369.61 kg/m³.

2.2 | Carr's index

The flowability of a product can be characterized by the Carr's index (CI), which is the compressibility index between the tapped density (TD) and the bulk density (BD); in other words, it describes the relative change in volume between the bulk volume and the tapped volume^{17,18} and can be expressed as

$$CI = \frac{(TD - TB)}{TD} \cdot 100\%. \quad (1)$$

According to Carr,¹⁹ larger Carr's index values (CI > 25%) indicate poor flowability, while Carr's index values smaller than 15% indicate good flowability. The Carr's index in this case was 13.3%.

2.3 | Angle of repose

The angle of repose (AR) describes the angle of inclination between a horizontal plane and the surface of a conical heap of loose, uncompressed bulk material (in static equilibrium)¹⁷; Figure 4 shows the angle of repose of sand. The angle of repose is the height (H) and radius (R) of the pile:

$$AR = \tan^{-1} \frac{H}{R}. \quad (2)$$

The angle of repose of the product coffee is 46.5°.



FIGURE 4 Exemplary angle of repose/slope angle of coarse sand²⁰

2.4 | Shear properties

The characteristic shear properties are given by the flow function and the compressive strength, which relates to the cohesiveness of a powder.^{17,21,22} In general, flow functions characterize the solidification behaviour of bulk solids and represent their flow limits. In bulk solids mechanics, a flow limit is referred to as a flow point^{17,23–26} (Figure 5) and describes the force that has to be applied in the form of shear stress τ in order to make the bulk solids flow. This knowledge is necessary in order to design a silo in such a way that no flow problems occur (e.g., no flow of bulk material due to bridging).⁷ The location of the flow location depends on the previous solidification. This means that the more solid a bulk material is, the greater the density and strength of the bulk material, and greater tension must be applied in order to induce the bulk material to flow. Exactly one bulk density and one flow location can be found for every consolidation stress. The flow function can be represented by measuring several flow locations. The determination of the flow location was carried out with the RSV95 ring shear device (Figure 6). The bulk material sample is poured into a ring-shaped shear cell, and a lid adapted to the shape of the shear cell with a rough underside is placed on the sample, via which the normal stress is applied to the sample (Figure 6). The bulk material is sheared by turning the lid and shear cell relative to 1. The torque required for this purpose is measured and converted into a shear stress. The resulting flow function for the coffee sample product used is shown in Figure 5. The consolidation stress σ_1 was 2281 Pa, the compressive stress σ_c 1073 Pa, the effective friction angle φ_e (which is a measure used for evaluating the inner friction in stationary flow) to 48°, the internal friction angle in stationary flow φ_{sf} to 45°, which is often used instead of the inner friction angle (defined as the local inclination of the point of flow against the σ axis), inclination φ_{lin} of the linearized point of flow against the σ -axis to 35°, and the wall friction angle φ_w to 39°, as well as the ratio ff_c of consolidation stress

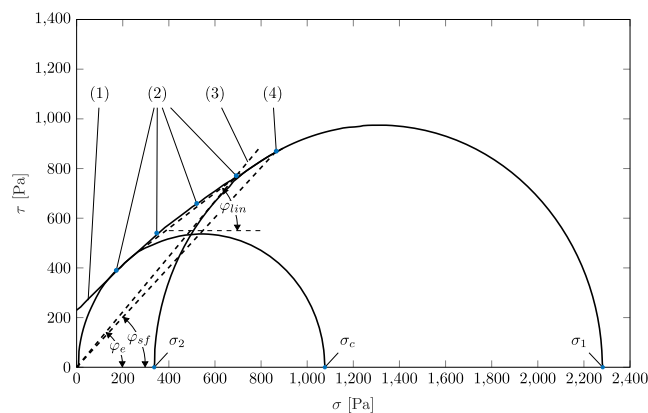


FIGURE 5 Flow function of the product coffee: (1) flow limit or flow point, (2) shearing points (beginning to flow), (3) effective flow point, (4) shearing point (steady flow), σ_c compressive stress, σ_1 consolidation stress, σ_2 horizontal main stress, φ_e effective friction angle, φ_{sf} internal friction angle in stationary flow, φ_{lin} inclination of the linearized point of flow against the σ -axis

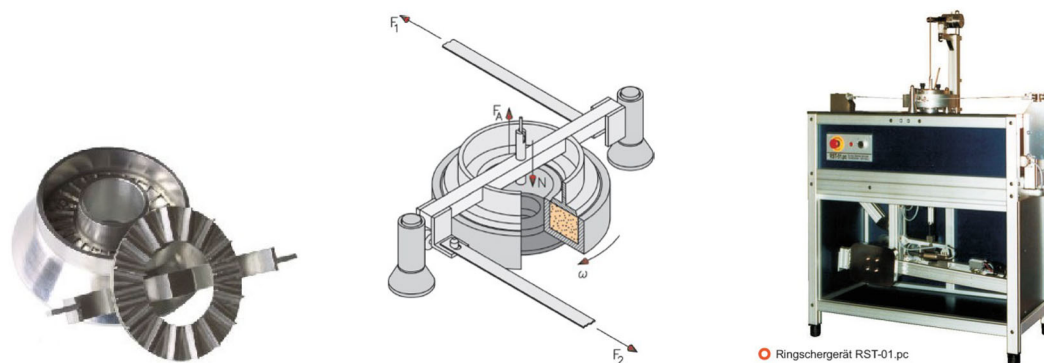


FIGURE 6 Shear cell (left), complex structure of the shear cell (middle) and test stand for determining shear properties (right)²⁷

σ_1 to compressive stress σ_c , which is often used to characterize the flow behaviour of bulk materials to 2.1 (Figure 5). In general, the greater the ratio ff_c , the better a bulk material flow. Powders with ff_c below 4 denote poor flow.¹⁷ ff_c values greater than 10 denote good flow behaviour of the powder.¹⁷

3 | EXPERIMENTAL TEST SET-UP

In order to increase the dosing accuracy of a vertical screw dosing device by controlling the screw on a predetermined volume or mass flow rate (at a constant product density, the volume flow rate is linearly dependent on mass flow rate and vice versa), a suitable sensor for determining the product density must be integrated. In the following, the sensor measurement principle is considered before the experimental set-up is shown as an example.

3.1 | Measuring principle for determining the product density

The actual mass flow can be determined by means of the product density and the screw volume as a function of the speed. The known possible measuring principles for contactless product density measurement are based on

- electromagnetic radiation/radiometry,
- change in electrical capacitance, and
- electromagnetic waves/microwave.

From previous investigations,²⁸ the measuring principle using electromagnetic waves/microwave technology has proven to be particularly suitable, which is why it is used in the following. The aim is to show that the mass flow control system is an improvement on conventional methods.

An SLH-46 microwave sensor is used as a sensor in conjunction with the MW4260 control unit from TEWS Elektronik GmbH & Co. KG. This microwave sensor can be used to determine the product density and moisture content of a powdery bulk material in a

tube with an internal diameter of 45.5 mm. The measuring principle of the sensor is generally based on the resonance method. By passing the bulk material or powder to be measured through an electromagnetic field, the resonance of the electromagnetic field is measured and evaluated. The product or the water molecules of the product cause a frequency shift and an attenuation of the amplitude. The quotient of the frequency shift and the amplitude attenuation is used for the moisture measurement. For density measurement, however, either the frequency shift or the amplitude attenuation can be used. Both values are independent here. The frequency shift is used in the following measurements to determine the product density value, since the shift range is larger than the small change in the attenuation.

3.2 | Experimental set-up

Figure 7 shows the exemplary test set-up. The auger filler FVS 3111 from Syntegon Technology GmbH (formerly Robert Bosch Packaging Technology GmbH) was used as the filler. In the first experiment, the microwave sensor is integrated directly at the product discharge point—that is, at the end of the dosing screw. In a further experiment, the sensor is integrated directly after the filling funnel. Due to the sensor used and the measuring principle, the dosing screw and the dosing tube are made from a high-molecular thermoplastic material—polyoxymethylene (POM). POM was selected for its mechanical properties, in particular, its shear stress and strength, as well as its excellent sliding and wear behaviour. In addition, POM is suitable and authorized for use in contact with food. In order to achieve a torsional stiffness similar to that of a stainless-steel screw, the diameter of the screw core, also known as a chord, was increased from 7 to 17 mm.

3.3 | Sensor calibration

The microwave sensor used can be operated in two sensor modes—either with high sensitivity or low sensitivity. The sensitivity generally reflects the sensitivity of the sensor to the product. The

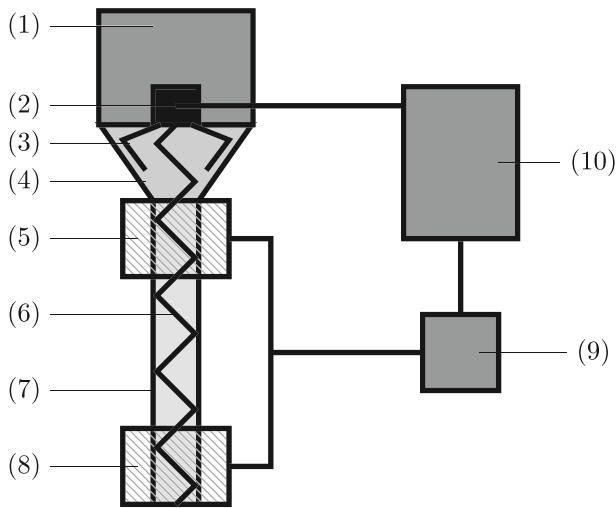


FIGURE 7 Exemplary structure of the test stand used: (1) filling container, (2) motor of the dosing screw, (3) stirring device, (4) filling funnel, (5) microwave sensor (installation in the first experiment), (6) dosing screw, (7) dosing tube, (8) microwave sensor (installation in second experiment), (9) control unit of the sensor, (10) machine control

lower the sensitivity, the less sensitive the sensor is to interferences since a weaker field is used for measurement. In the case of powdery bulk goods with a high product density, when the low-sensitivity mode is used, the attenuation of the emitted electromagnetic field, that is, the frequency shift and attenuation of the amplitude, can be so great that no density or measured value can be determined from it. In such cases, the sensor mode with higher sensitivity must be used. The sensor mode with higher sensitivity was used for the coffee sample product.

As described above, the frequency shift is used to determine the density value. However, this is dependent on the product, which requires calibration of the sensor. A semiautonomous process was developed for this. The aim of this process is to produce as different density values as possible by modifying various process parameters in order to cover the entire range of density values occurring via the resulting calibration curve. By varying the stirrer speed from 0 to 100% at a screw speed of 1600 rpm and the screw speed in the range from 400 to 1600 rpm at a constant stirrer speed of 40% as well as different filling levels of the filling container, several density values were produced and recorded by means of a sensor. For this purpose, the exact volume of product in the sensor area was measured. From the measuring range of 246 mm and the screw pitch of 50 mm, a 4.92 number of revolutions must be filled. The mean measured sensor value is compared with the actual determined density value (Figure 8). Here, the density value was determined by measuring the mass using a laboratory balance—an FH-2000 model from GRAM—and the actual net displacement volume V_N according to Hohmann⁴ and Vogelpohl and Hohmann.⁵ The net displacement for one revolution is calculated by subtracting the volume of the tube by the volume that the screw displaces and is shown in Equation (3). D_1 is the internal diameter of

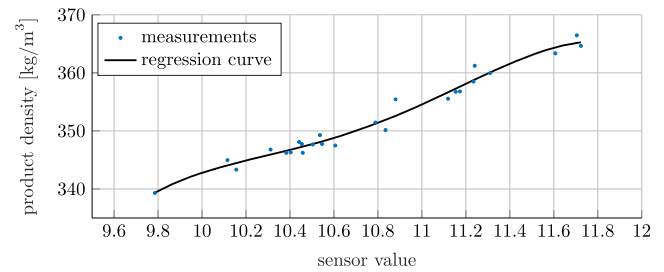


FIGURE 8 Determined calibration function of the sensor for the used sample product coffee; calibration function: $C(S) = -6.194 \cdot S^4 + 266 \cdot S^3 - 4276.3 \cdot S^2 + 30\,515 \cdot S - 81\,231$

the metering tube, which is 45.5 mm, while D_2 is the core diameter of the screw (17 mm), and b is the thickness of the screw blade (1.2 mm). The pitch s is also referred to as the screw pitch, which in this case is 50 mm.

$$V_N = \left(\frac{D_1^2 - D_2^2}{4} \right) \cdot \pi \cdot s - \left(\frac{D_1 - D_2}{2} \right) \cdot b \cdot \sqrt{\left(\frac{D_1 + D_2}{2} \right)^2 \cdot \pi^2 \cdot s^2} \quad (3)$$

For the example product coffee, the determined sensor values lie in a coarse range between 9.5 and 12 (Figure 8). The measured values are transmitted to the machine control of the feeder via an analogue output. The measured value 6 is set as the lower limit of the output current of 4 mA, while the upper limit of the output, 20 mA, is set to the measured value 12. The measured values are read to the machine control by means of an analogue input card with a 12-bit resolution, which means that the transmitted current values are recalculated. The actual density value is determined with the aid of the back calculation in Equation 4 and the calibration function (Figure 8). S is the measured sensor value, and E is the voltage value on the input board of the machine control.

$$S(E) = 0.375 \text{ mA}^{-1} \cdot \left(\frac{E}{2048} + 4 \text{ mA} \right) + 4.5 \quad (4)$$

The determined density values are only valid in the calibrated range for the sensor value. Incorrect density values are assumed for all values outside the defined range and thus also an incorrect mass flow rate, which leads to an unreproducible amount in the dosing.

4 | MASS FLOW CONTROL

Knowing the density and the net displacement as a function of the speed makes it possible to determine the mass flow and to regulate it to a predetermined trajectory value. Equation (5) shows the relationship between the mass flow rate $\frac{dm}{dt}$, the volume flow rate $\frac{dv}{dt}$, and the density $\rho(t)$. The volume flow depends on the cross-sectional area and the flow velocity: in other words, on the volume V of one revolution multiplied by the speed n .

$$\frac{dm}{dt} = \frac{dv}{dt} \cdot \rho(t) = V \cdot n \cdot \rho(t) \quad (5)$$

In the following, a trajectory planning for the specification of the mass flow is shown first before the controller draft is then discussed.

4.1 | Trajectory planning of the given mass flow

From the specification of the dosing time t_D and the mass m_s to be dosed, it is possible to determine the necessary constant mass flow (mass m_s /dosing time t_D). At the beginning of dosing, the predetermined mass flow of 0 kg/s jumps to the determined constant mass flow. Such jumps can be made by real systems or by those whose actuators (in this case, the motor of the metering screw) are not realized—for this, the motor would have to jump from a standstill to the determined speed (speed), which it is unable to do. The result of a jump in the mass flow would indicate that the motor is unable to follow the mass flow specification and the associated specification of the rotational speed (condition of a constant density), which results in a deviation from the regulation and leads to overshoot. On the other hand, the metering screw is accelerated to the speed (speed) with the maximum possible acceleration, whereby the motor and the metering screw are heavily stressed. In order to minimize the operation of the servomotor and the control deviation in the acceleration and braking phase of the motor in a gentler manner, the mass flow in these phases is linearly increased or reduced. The resulting predetermined trajectory of the mass flow resembles a trapezoidal shape (Figure 9) and is calculated from functions of the first order (Equation 6). The dosing time for starting up x_1 and braking x_2 of the servo motor are given as a percentage and may not exceed 0.5 (corresponding to 50% of the dosing time t_D). The specified mass flow rate $\frac{dm_s}{dt}$ results in

$$\frac{dm_s(t)}{dt} = \begin{cases} \frac{\dot{m}_{Max}}{x_1 \cdot t_D} \cdot t, & t \leq x_1 \cdot t_D \\ \dot{m}_{Max}, & x_1 \cdot t_D \leq t \leq (1-x_2) \cdot t_D \\ \frac{\dot{m}_{Max}}{x_2 \cdot t_D} \cdot (t - ((1-x_2) \cdot t_D)), & t > (1-x_2) \cdot t_D \end{cases} \quad (6)$$

with $\dot{m}_{Max} = \frac{m}{\frac{(x_1+x_2) \cdot t_D}{2} + ((1-x_1-x_2) \cdot t_D)}$

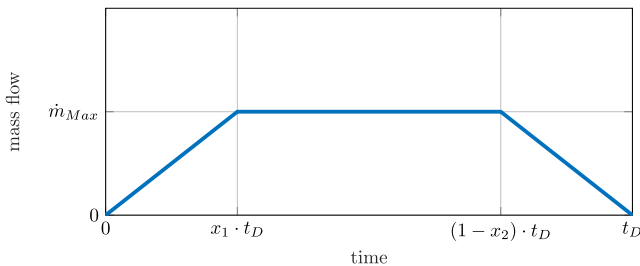


FIGURE 9 Predetermined calculated mass flow as a function of the quantity m_s to be dosed and the dosing time t_D as well as a percentage indication of the dosing time for starting up x_1 and braking x_2 of the servomotor based on Equation 6

where $\dot{m}_{Max}$ is the maximum calculated mass flow.

When changing a product or altering the predetermined mass flow by adjusting a parameter, it is first checked whether the quantity to be metered (predetermined mass flow) is below the given process parameters (motor parameters and product density) in the time to be dosed. The density value is determined over 10 revolutions and assumed to be constant.

4.2 | Control unit design

Numerous approaches to control methods are known in control engineering and system dynamics. In many applications, a conventional control loop—consisting of a system and a controller—is extended with a feedforward control in order to achieve a better control quality. The inverse of the transfer function²⁴ is the ideal choice for the pilot control, which is why these pilot controls are also referred to as model-based pilot controls. Model-based pilot controls, also known in the literature as pilot control with a model control loop, are used^{29–31} to set the control behaviour of a control system, which is why this control loop approach known from the literature is used (see Figure 10).

Using this feedforward control has the advantage that disturbances are better controlled, since the controller can be adjusted more finely and only needs to correct uncertainties and disturbances in the model.³¹ In addition, the dynamics are improved without influencing the stability and an additional degree of freedom is created for optimizing control and disturbance behaviour.³¹ Furthermore, the control and the pre-control can be designed separately from each other in the linear case.³¹

The structure of the model control loop used is shown as an example in Figure 10, where G is the system, G_M is the system model, and G_{MR} , G_R are the controller of the feedforward and the controller. A PI controller is used as the actual controller. A PI controller is medium-fast and regulates permanent control deviations, but this should be sufficient for the measurement task. The controller is designed according to the Ziegler and Nichols method, with the adjustment to the stability margin. The model used here can be described by the following equation, where $\sum M$ is the sum of all torques—frictional torque, engine torque, etc.— J is the moment of inertia, and $\ddot{\varphi}$ is the angular acceleration, that is, the derivative of the angular velocity and thus the engine speed.

$$\sum M = J \cdot \ddot{\varphi} \quad (7)$$

In simplified terms, the system behaviour of the servomotor can be assumed as PT1 transmission behaviour for the controller design. At a maximum nominal speed of 2000 rpm, this results from a time constant T of 10 so that the motor reaches its nominal speed in about 50 ms. The control parameters of the feedforward controller resulted from $K_{pF} = 0.3$ and $K_{iF} = 0.1$ and those of the following controller to $K_p = 0.15$ and $K_i = 0.03$.

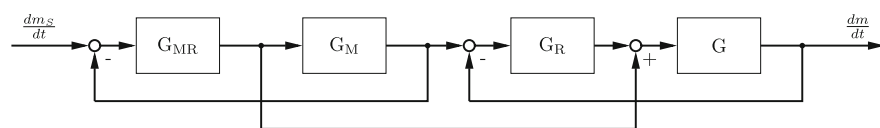


FIGURE 10 Used control: model-based control according to Wurmthaler and Kühnlein³⁰; G_{MR} feed forward controller, G_M system model, G_R controller, G system

5 | EXPERIMENTS AND DISCUSSION

5.1 | Experiment 1: Measurement of the mass flow at the product delivery point

In this experiment, the microwave sensor for density measurement is integrated directly at the product delivery point (see Figure 7). The dosage amount has been set to 125 g and 250 g and the dosing time to 500 ms, 650 ms, and 800 ms. The results of mass flow control by density measurement are shown in Table 2. In each case, three series of measurements with 20 individual doses per dosing time and dosing weight were dosed and the resulting mean value and mean deviation were calculated.

The results from Table 2 show an increase in reproducible dosing accuracy with increasing dosing time or, in other words, the longer the dosing time for a dosing quantity, the lower the standard deviation. Due to the longer dosing time, more density measurement values are also used for the mass flow control, whereby the measured mass flow rate of the changing product can be more accurately determined in the chronological sequence.

5.1.1 | Comparison with the conventional control

In the conventional control, the most suitable dosing parameters (the target speed and the screw rotations) are determined iteratively depending on the product and the quantity to be dosed, and the dosing device is specified as a target parameter. With many products, the standard deviation decreases with increasing dosing time, which is why the dosing time is often maximized. Powdery bulk materials are often packed by means of vertical form, fill and sealing machines, whereby the capacity utilization is often 60 bags per minute. This results in a maximum dosing time of 1000 ms. Due to the fluctuation of the product density during operation, the number of screw rotations is continuously adjusted by means of statistical methods, whereby the dosing time changes continuously and is not constant. The speed and revolutions of the dosing screw were set in the conventional control so that the dosing time is about 800 ms. Due to the continuous adjustment of the revolutions and the resulting different dosing time (maximum 1000 ms), the dosing time cannot be set to a value, which is why it is specified below 1000 ms. In Table 3, the values of the conventional control with the lowest standard deviation, including the lowest deviation of the mean to the net weight, are compared with the measured values of the mass flow control with the worst standard deviation.

The results from Table 3 show the highest standard deviation of 1.24 g in the worst case at a dosing time of 800 ms and a dosing weight of 250 g—however, this value is an outlier. If the outlier is

not considered, in the worst case, the mass flow control can save 0.55 g of product per dosage, resulting in a time saving of around 37% compared with a conventional control. If the mean value is considered, the measurement results show an underfill at a dosing time of 650 ms and therefore does not comply with the legal requirements—for example, the prepackaging regulation in Germany.⁶ This underfill is due in part to the sensor calibration curve. The calibration curve was determined as described above by changing the settings of the feeder and averaging the density value. This curve shows a different value for some density values, which is related to the flowability of the product. In general, the deviation from the net weight depends on the calibration accuracy of the sensor. Furthermore, there is a presumption that micro-bridges form, leading to a prevention of the product falling as waste. An attempt was made to analyse this with a high-speed camera, but this did not provide any clear evidence for or against this assumption. Viewed over all three-measurement series, the underfilling of the worst case under consideration has compensated for itself and the net weight has been complied with.

5.2 | Experiment 2: Measurement of the mass flow at the beginning of the dosing tube

Due to the size of the sensor, it is not always possible to integrate the sensor at the product delivery point, for example when dosing in vertical form, fill and sealing machines, which is why the sensor is integrated at the beginning of the dosing tube in this experiment. The density values were then determined. With the aid of information about the screw speed and the distance to the product delivery point, the dead time can be calculated and thus the actual measured value for the product currently being delivered can be determined. It is assumed here that the product does not change in the screw. In this experiment, the same dosing times as in the first experiment were investigated. The results of 20 doses for the different dosing times are shown in Table 4.

With increasing dosing time, the standard deviation decreases: This means that the reproducibility increases, as already seen in the first experiment. For the measurements carried out, the standard deviation is at the level of the first experiment. However, a significant overcrowding is noticeable. The assumption of the test carried out here is based on a constant product behaviour in the dosing tube during the vertical conveyance to the delivery point. On the one hand, it is possible that the flowability and the conveyance of the product result in compression. On the other hand, micro-bridges can arise during conveyance, which can lead to product dropping at the product delivery point. Furthermore, the possible calculation inaccuracy can also play a key role in. The program is set cyclically in a fixed task. By

TABLE 2 Measurement results of bottled coffee by mass flow control of a screw feeder; measurement of the mass flow at the product delivery point

Mass flow control						
Dosing time	500 ms		650 ms		800 ms	
Dosing weight	125 g	250 g	125 g	250 g	125 g	250 g
Mean value	125.15 g	250.17 g	125.04 g	250.00 g	125.18 g	250.85 g
Standard deviation	0.75 g	1.15 g	0.43 g	0.97 g	0.37 g	0.92 g

TABLE 3 Comparison of the control in the worst case—conventional control with the lowest standard deviation and mass flow control with the worst standard deviation; measurement of the mass flow at the product delivery point

Conventional control			Mass flow control				
Dosing time	<1000 ms		500 ms		650 ms		800 ms
Dosing weight	125 g	250 g	125 g	250 g	125 g	250 g	125 g 250 g
Mean value	128.41 g	252.27 g	125.29 g	250.07 g	124.55 g	248.81 g	125.83 g 251.28 g
Standard deviation	1.89 g	1.79 g	0.78 g	1.17 g	0.50 g	1.11 g	0.40 g 1.24 g

TABLE 4 Measurement results on bottled coffee by means of mass flow control of the screw feeder; measurement of the mass flow at the beginning of the dosing tube

Mass flow control						
Dosing time	500 ms		650 ms		800 ms	
Dosing weight	125 g	250 g	125 g	250 g	125 g	250 g
Mean value	-	256.01 g	-	256.24 g	-	256.38 g
Standard deviation	-	1.12 g	-	0.96 g	-	0.69 g

TABLE 5 Comparison of the control in the worst case—conventional control with the lowest standard deviation and mass flow control with the worst standard deviation; measurement of the mass flow at the beginning of the dosing tube

Conventional control			Mass flow control				
Dosing time	<1000 ms		500 ms		650 ms		800 ms
Dosing weight	125 g	250 g	125 g	250 g	125 g	250 g	125 g 250 g
Mean value	128.41 g	252.27 g	-	255.20 g	-	256.05 g	- 256.38 g
Standard deviation	1.89 g	1.79 g	-	1.21 g	-	1.04 g	- 0.88 g

altering and adapting the speed, the dead time until the measured product is dispensed also changes. The fixed cycle time means that the dead time of the measured value is between two cycles. The preceding and following values are also included as a percentage, which leads to a maximum error of 6.28 g at a maximum speed of 1600 rpm and an assumed density of 345 kg/m³. In summary, it can be concluded that there is continuous overfilling, with an average of 6.38 g of coffee for the example product coffee used. One possibility is to minimize the net quantity by the determined difference. There is also the option of continuously adjusting the target weight using statistical methods in order to minimize further overfilling.

5.2.1 | Comparison with the conventional control

In Table 5, the measured values of the mass flow control (installation of the sensor at the beginning of the dosing tube) are compared with the results of the conventional control. Here again, the worst-case

scenario was considered. For conventional control, the lowest deviation series was used, and for the mass flow control, the highest standard deviation series.

In the worst case, the standard deviation is reduced by 0.58 g when using a mass flow control instead of the conventional control. In addition, there is also a significant improvement in the reproducibility of the individual dosages and even a reduction in the dosing time by around 37%. In summary, 0.58 g per dosage could be saved with a time saving of around 37% by means of mass flow control.

6 | CONCLUSION

The measuring principle used has the advantage of using the direct feedback via the density measurement of the product and thereby lowering the standard deviation, which results in a significant reduction of the dosed weight. In the present case, the use of the mass flow control can save 0.55 g of coffee per dosage and still reduce

the dosing time by around 37%—this saves the manufacturer a lot of money, and all legal regulations are complied with. However, the deviation from the net or target weight is highly dependent on the sensor calibration and also on the flowability of the product and, in particular, on the product behaviour during the conveyance in the dosing screw to the product delivery point. Due to the limited volume of the filling container or the small silo used, no more than 20 individual doses were possible, which is why this has to be checked in continuous operation in an industrial environment. Furthermore, the principle must be verified with other products. Initial measurements have shown that a significant increase in dosing accuracy can be achieved with even more flowable products such as flour or corn starch.

ACKNOWLEDGEMENTS

Open Access funding enabled and organized by Projekt DEAL.

DATA AVAILABILITY STATEMENT

The main data are mentioned in the article.

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How to cite this article: Kruppa F, Weiß U, Oberdorfer B, Wilke B. Increasing the dosing accuracy of a screw dosing device by inline measurement of the product density. *Packag Technol Sci.* 2023;36(3):185-194. doi:10.1002/pts.2703